Tool 1:- Multiple regression using python

In [1]:
```
from sklearn.linear_model import LinearRegression
import pandas as pd

```

In [4]:
```
data = pd.read_excel('calculations.xlsx')
```

In [5]:
```
data
```

Out[5]:

	danceability_%	valence_%	energy_%	acousticness_%	instrumentalness_%	liveness_%	speechiness_%
0	80	89	83	31	0	8	4
1	71	61	74	7	0	10	4
2	51	32	53	17	0	31	6
3	55	58	72	11	0	11	15
4	65	23	80	14	63	11	6
...	...	...	...	...	...	...	...
948	60	24	39	57	0	8	3
949	42	7	24	83	1	12	6
950	80	81	67	4	0	8	6
951	82	67	77	8	0	12	5
952	61	32	67	15	0	11	5

953 rows × 7 columns

In [10]:

```python
import pandas as pd
import statsmodels.api as sm

df = pd.DataFrame(data)

# Add a constant for the intercept term
df['intercept'] = 1.0

# Specify independent variables
X = df[['valence_%', 'energy_%', 'acousticness_%', 'instrumentalness_%', 'liveness_%', 'speechiness_%', 'intercept']]

# Specify dependent variable
y = df['danceability_%']

# Fit the regression model
model = sm.OLS(y, X).fit()

# Print the regression results
print(model.summary())

```

THIS IS THE CODE USED TO CALCULATE MULTIPLE REGRESSION .

OUTPUT:-

```
==============================================================================
Dep. Variable:         danceability_%   R-squared:                       0.248
Model:                            OLS   Adj. R-squared:                  0.243
Method:                 Least Squares   F-statistic:                     52.01
Date:                Thu, 21 Dec 2023   Prob (F-statistic):           1.93e-55
Time:                        09:28:52   Log-Likelihood:                -3772.9
No. Observations:                 953   AIC:                             7560.
Df Residuals:                     946   BIC:                             7594.
Df Model:                           6
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
valence_%             0.2559      0.019     13.281      0.000       0.218       0.294
energy_%             -0.0736      0.033     -2.211      0.027      -0.139      -0.008
acousticness_%       -0.1410      0.020     -7.144      0.000      -0.180      -0.102
instrumentalness_%   -0.0309      0.050     -0.622      0.534      -0.129       0.067
liveness_%           -0.0921      0.030     -3.035      0.002      -0.152      -0.033
speechiness_%         0.2341      0.042      5.598      0.000       0.152       0.316
intercept            61.7085      2.422     25.483      0.000      56.956      66.461
==============================================================================
Omnibus:                       26.901   Durbin-Watson:                   1.919
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               22.073
Skew:                          -0.295   Prob(JB):                     1.61e-05
Kurtosis:                       2.544   Cond. No.                         537.
==============================================================================
```

Here the model suggests that there is some relationship between the independent variables and the dependent variables i.e. danceability_%.

Tool 2:- Correlation using python:-

In [1]:
```
import pandas as pd
```

In [4]:
```
data = pd.read_excel('correlation.xlsx')
```

In [5]:
```
data
```

Out[5]:

	danceability_%	valence_%
0	80	89
1	71	61
2	51	32
3	55	58
4	65	23
...	...	...
948	60	24
949	42	7
950	80	81
951	82	67
952	61	32

953 rows × 2 columns

In [6]:
```
data.head()
```

Out[6]:

	danceability_%	valence_%
0	80	89
1	71	61
2	51	32
3	55	58
4	65	23

In [3]:
```
# Calculate the correlation matrix
correlation_matrix = data.corr()

# Print or display the correlation matrix
print(correlation_matrix)
```

Output:-

	danceability_%	valence_%
danceability_%	1.000000	0.408451
valence_%	0.408451	1.000000

Hence, There is a moderate positive correlation between danceability_% and valence_% also increase in danceability_% is associated with an increase in valence_%

Tool 3:- z-test two sample for means using excel:-

Setting the hypothesis:-

1. null hypothesis(H0):- There is no difference between the streams of single and multiple artists.

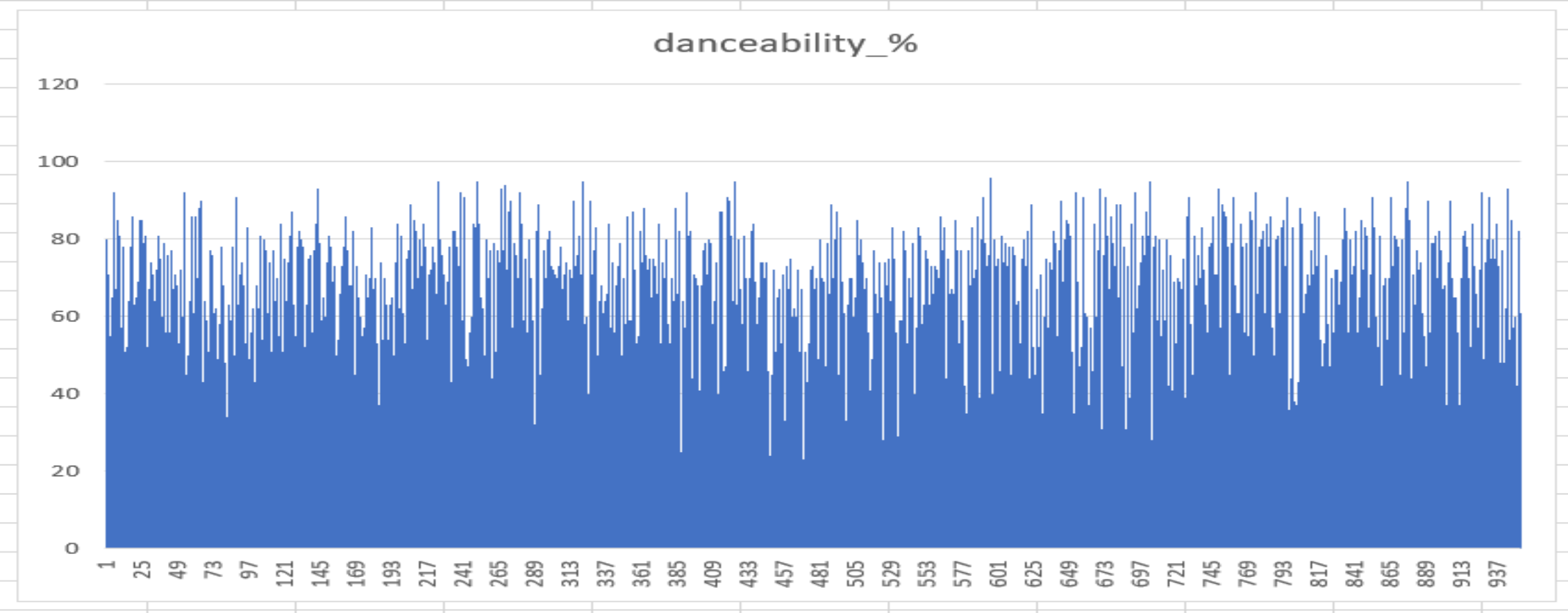
2. Alternative hypothesis(H1):- There is a difference between the streams of single and multiple artists.

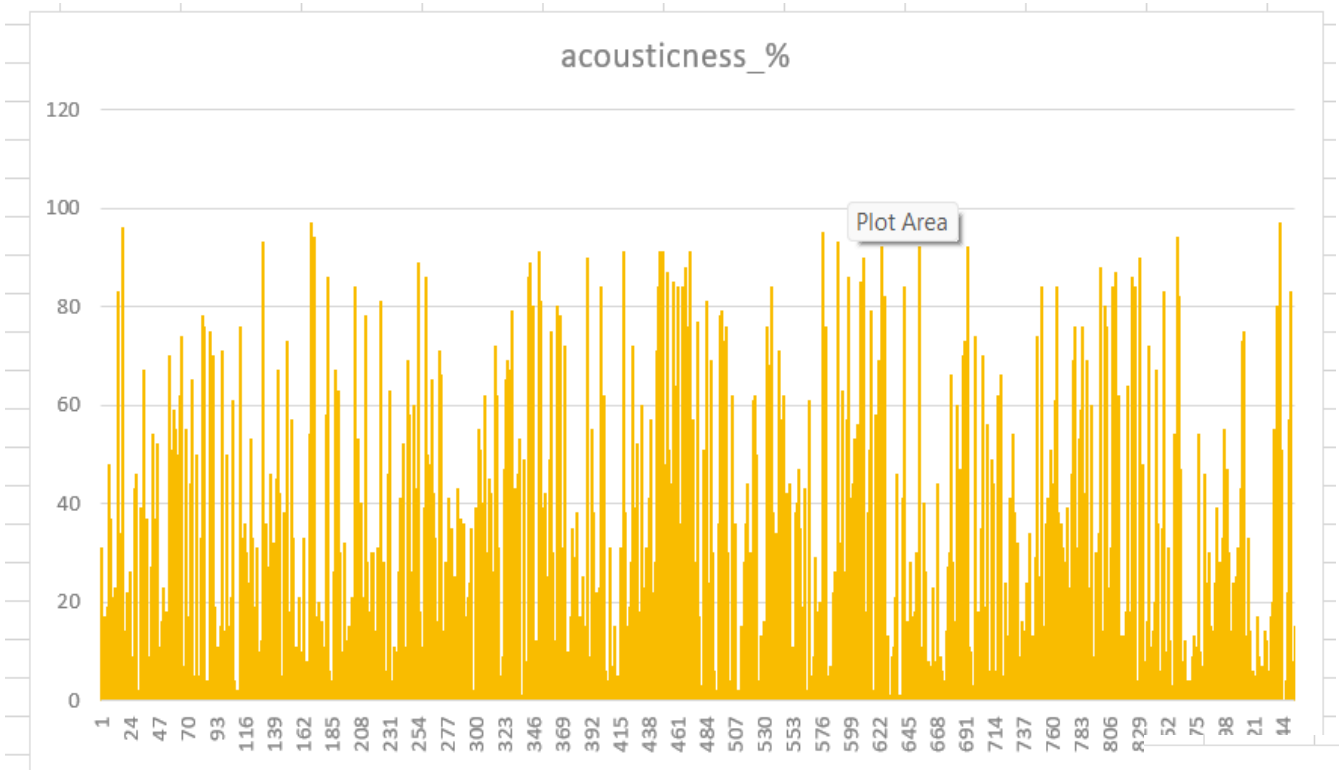
OUTPUT:-

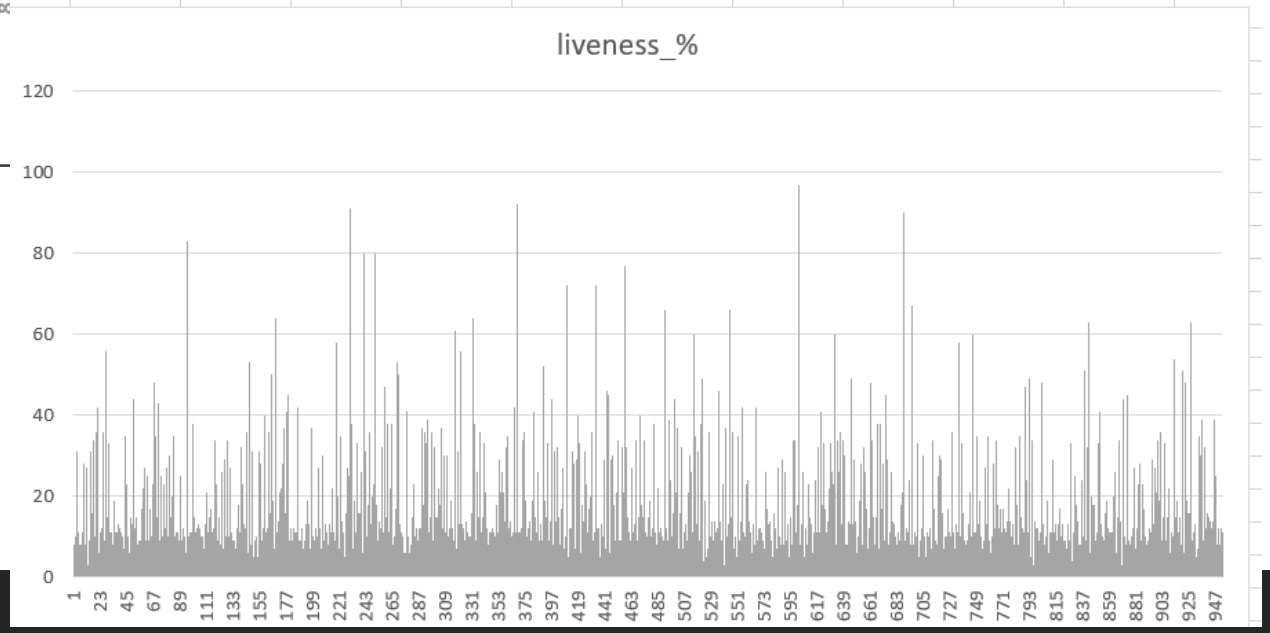
z-Test: Two Sample for Means		
	Variable 1	*Variable 2*
Mean	568208266.1	427559547.8
Known Variance	3.51835E+17	2.60136E+17
Observations	587	366
Hypothesized Mean Difference	0	
z	3.885779951	
P(Z<=z) one-tail	5.10009E-05	
z Critical one-tail	1.644853627	
P(Z<=z) two-tail	0.000102002	
z Critical two-tail	1.959963985	

- The calculated z-statistic is 3.885779951, and both the one-tailed and two-tailed p-values are very small (close to zero).
- Since the calculated z-statistic is greater than the critical values for both one-tailed and two-tailed tests, you would likely reject the null hypothesis.
- This suggests that there is a significant difference between the means of Variable 1 and Variable 2.
- That means there is a significant difference between streams of the single and multiple artists.
- Also mean of the streams of single artists are more as compare to the mean of the streams of multiple artists.

Charts:- Created bar charts for feature to show their distribution. This helps to understand the range and variability of the feature.




acousticness_%
120
100
80
60
40
20
0
Plot Area
1
24
47
70
93
116
139
162
185
208
231
254
277
300
323
346
369
392
415
438
461
484
507
530
553
576
599
622
645
668
691
714
737
760
783
806


liveness_%
120
100
80
60
40
20
0
1
23
45
67
89
111
133
155
177
199
221
243
265
287
309
331
353
375
397
419
441
463
485
507
529
551
573
595
617
639
661
683
705
727
749
771
793
815
837
859
881
903
925
947